

General information

Lecturers

- Dr. Mario Gellrich (module responsible)
- Dr. Alexander Walter

Documents

- All course materials can be found on the electronic learning platform "Moodle" and on GitHub.

Tools and software

- Generally, we will use GitHub Codespaces, Python and Jupyter notebooks for all exercises.
- Installation instructions for additional tools will be provided or linked on Moodle.

Location / teaching format

Large classes (German: Grossklassen, GK) are organized as follows:

- All lectures will be held on-site in Winterthur (see WebUntis for rooms and schedules).

Small classes (German: Kleinklassen, KK) are organized as follows:

- In the first week of the semester, all small classes will be held on-site in Winterthur (see WebUntis for rooms and schedules).
- From the second week of the semester onwards, small classes will take place online via [MS Teams](#).
- Note that all online small classes can be attended by every student at any time!

Changes in the location and form of instruction will be announced in a timely manner.

Weekly exercises

The weekly exercises in the small classes must **not** be submitted on Moodle; they serve as preparation for the written exam.

End of module exam / final grade

Only the current module description is relevant with regard to the exam and composition of the final grade.

The final grade is determined as:

- 60% grade of written exam (regular end-of-module exam, MEP)
- 40% grade of project work with the two parts (each 20%):
 - group presentation of the project concept (for details see below)
 - final project submission on Moodle (for details see below)

The written exam will ...

- last 60 minutes
- be open book
- cover the entire material of the module
- contain single-choice questions (4 possible answers per question, only one is correct).

- **Programming environment & recommended subscriptions**

In this course, we will mainly work with Python and Visual Studio Code using GitHub Codespaces.

Therefore, you need a **GitHub Pro account**, which you can apply for via ‘educational benefits’ on GitHub.

After you have created your own GitHub Pro account, you can create a Codespace from this repository:

https://github.com/mario-gellrich-zhaw/data_analytics.git

We recommend setting up a payment method and limiting the amount to **10 U.S. dollars per month**.

Although AI models (LLMs) are part of the GitHub ecosystem, the number of AI credits per month is limited. We therefore additionally recommend purchasing either a **Copilot Pro** or a **Claude Pro** subscription.

- **Evaluation of the project concept presentation**

Presentation Outline:

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- The ***presentations of the project concept*** will be on-site in Winterthur and take place at the beginning of the lectures (large classes) from week 10 onwards (3-4 groups per lecture, 10 min. per group).
 - For the presentation, provide a slide set (PDF-Document) and upload it to Moodle (see link below).
 - A link to the presentation dates/times is available on Moodle (same link as for the group registration).
 - Note that the dates/times of the presentations will be determined by the lecturer once the project groups are registered.
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To be evaluated:

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5. **Structure & outline** (Clear outline presented and followed? Logical flow with recognizable thread? Balanced weighting of parts? Title clear? Names of all students available?)
 6. **Background & problem** (Context of the project clear? Problem convincingly motivated?)
 7. **Objective & research question** (Clearly defined?)
 8. **Data** (Source shown? Structure of the data explained?)
 9. **Methods** (Methods appropriate for the research question? Correctly described and justified?)
 10. **Expected results** (Plausible? Linked back to the research question?)
 11. **Language** (Freely spoken?)
 12. **Quality of the presentation** (Quality of graphics, tables & slides according to scientific standards?)
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Grading scale (criteria 1–8)

Grade	Description
6	Excellent — the criterion is fulfilled completely and convincingly, without notable weaknesses
5	Good — the criterion is fulfilled well, with only minor weaknesses
4	Sufficient — the criterion is fulfilled in its essential aspects; basic requirements are met
3	Insufficient — key elements of the criterion are missing or incorrect
2	Poor — the criterion is addressed only in fragments
1	Not fulfilled — the criterion is not addressed / no performance shown

Each of the criteria 1–8 is graded independently on the scale 1–6. The final grade for the presentation of the project concept is the average of these eight grades.

- **Evaluation of the final project (max. 22 points)**

Minimum requirements (9 points)

The project must be a Data Analytics project, i.e., must include content of the Data Analytics module.

The project must integrate several topics presented in the module, at a minimum these are the following:

1. **Data collection** using Web Scraping and/or a Web API. The data must be collected by the group itself; ready-made datasets (e.g. from Kaggle) are not permitted as the primary data source.
 2. **Data preparation and integration:** the project must combine data from at least two different sources (e.g. scraped data enriched with data from opendataswiss). The integration step (matching keys, joining, resolving inconsistencies, number of records lost/gained) must be documented.
 3. **Data storage** in a SQLite database including SQL queries from within Python.
 4. **Rich non-graphical and graphical exploratory data analysis (EDA).**
 5. **Modeling:** use of either regression or classification. The research question must go beyond a standard prediction task (e.g. include a comparison between groups, regions, time periods, or data sources).
 6. **Model evaluation** using suitable measures of fit (e.g. r-squared, RMSE, accuracy, recall, precision, ...).
 7. **Correct interpretation** of model results and measures of fit.
 8. **Documentation of AI usage** for creating the code and the presentation (the use of AI is not only allowed but desired), including the advantages and disadvantages you found and where something did not work with the help of AI.
 9. **Making the material** (data, Jupyter notebooks, ...) available on Moodle.
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Note: The topic must be approved by the lecturers at the presentation of the project concept on-site in Winterthur (week 10 onwards). Overused standard topics (e.g. plain used-car or real-estate price prediction without an extended research question) will be graded lower on creativity.

Additional points, if the following criteria are met (max. 6 points)

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10. **Creativity of implementation** (creative is anything not specified in the lessons and exercises).
 11. **Use of a PostgreSQL or similar database** (no SQLite!) with, e.g. SQL joins and aggregations from within Python.
 12. **Integration and visualization of geographical data.**
 13. **Use of a Chi-squared test, analysis of variance (ANOVA), or correlation analysis** (each test must include a p-value).
 14. **Use of k-means clustering** in addition to the regression or classification model.
 15. **Code on a publicly available GitHub repository** (provide the GitHub project URL in the project presentation).
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Project presentation in the form of a video recording (max. 7 points)

To be evaluated:

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16. **Structure & outline**
 17. **Topic presentation** (Topic clearly defined? How convincingly presented?)
 18. **Content** (substance, depth, weighting of parts)
 19. **Language / auxiliary tools** (Freely spoken?)
 20. **Quality of the presentation & slides**
 21. **Reflection on AI usage:** the presentation (slides and video) must include which AI tools were used and how, the advantages and disadvantages you found when using AI, and where something did not work with the help of AI.
 22. **Time keeping:** length of presentation in minutes = number of students per group $\times$ 5 minutes.
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Notes:

→ Use the slide set (PDF) as the basis for the presentation.

→ Show the Python code & live demos only if it improves the understanding of methods and results.

→ Do not just scroll through the code in the Jupyter notebooks without explaining the code and results.

→ AI is a tool, not a substitute for understanding. The evaluation focuses on your ability to explain and justify your data, methods, and results.

To be submitted on Moodle with the group number by one student per group

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23. **Video recording** of the project presentation as mp4 file,
e.g.: video_recording_group_01.mp4
(Ensure that the video does not exceed the max. allowed file size on Moodle. If the video is too large, split it and upload separate parts, e.g.: video_recording_group_01_part_1.mp4, video_recording_group_01_part_2.mp4, ...)
24. **Zip file** with all data and your Jupyter notebook(s),
e.g.: materials_group_01.zip
25. **Presentation as PDF file** (e.g. presentation_group_01.pdf) with the following minimal structure:
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- Title and full name of each student
 - Introduction (background, problem, objective, research question)
 - Materials and methods (data collection method, content of data, exploratory data analysis methods, modelling methods)
 - Results & Discussion
 - Conclusions
 - Reflection on AI usage (which tools, for which tasks, which advantages and disadvantages you found, and where something did not work with the help of AI)
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IMPORTANT!!! In the appendix of the PDF document of the presentation, include what you think you did to earn the points. Use screenshots and code snippets to clearly demonstrate this. Note that the final evaluation will be made by the lecturer, not exclusively based on your listing!